Fundamentals of Big Data Analytics

Lecture 17 – Hadoop Distributed File System

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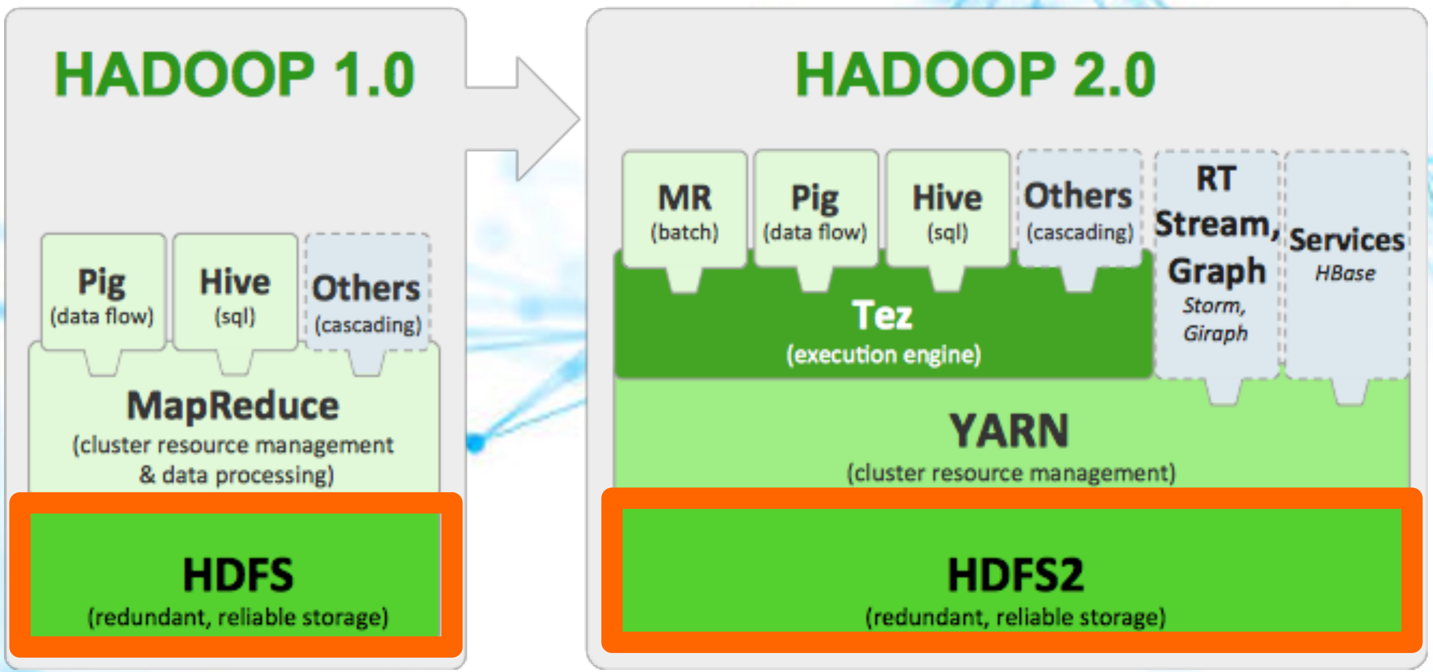


Apache Tez: A Framework for YARN-based, Data Processing Applications In Hadoop

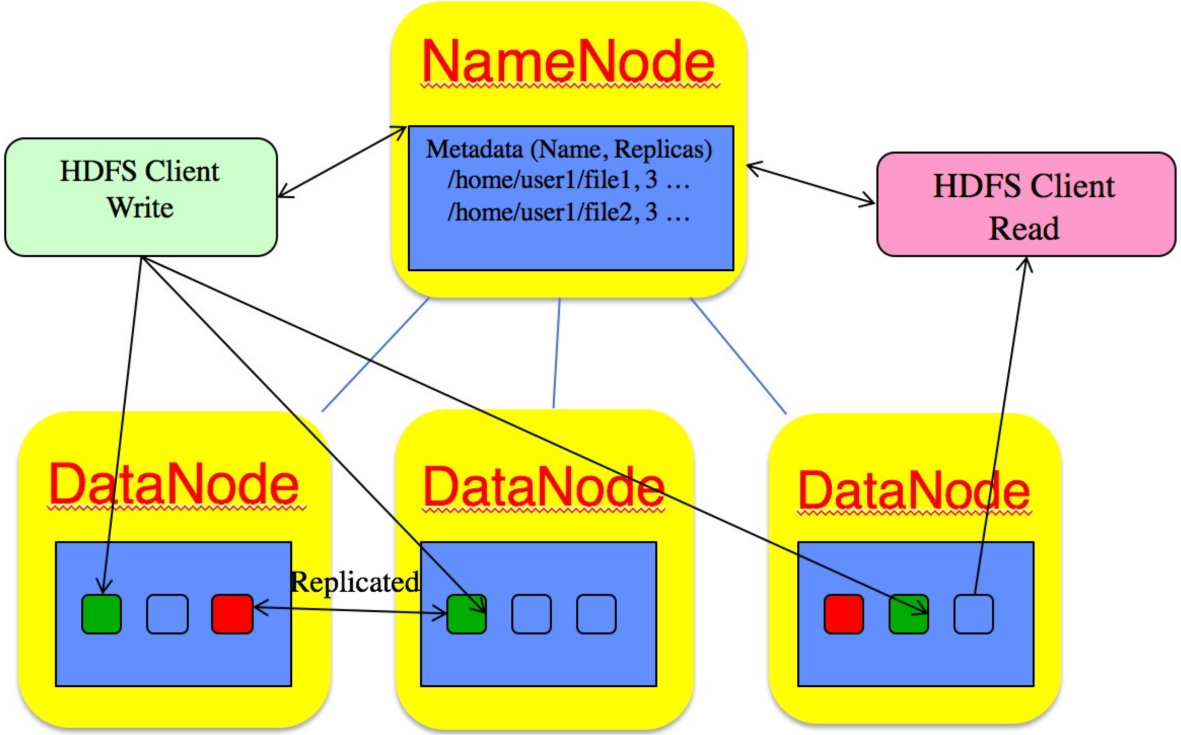
Apache™ Tez is an extensible framework for building high performance batch and interactive data processing applications, coordinated by YARN in Apache Hadoop. **Tez improves the MapReduce paradigm by dramatically improving its speed, while maintaining MapReduce's ability to scale to petabytes of data**. Important Hadoop ecosystem projects like Apache Hive and Apache Pig use Apache Tez, as do a growing number of third party data access applications developed for the broader Hadoop ecosystem.

- ***HBase* –** a scalable data warehouse with support for large tables.
- ***Hive* –** a data warehouse infrastructure that provides data summarization and ad hoc querying
- ***Pig* –** A high-level data-flow language and execution framework for parallel computation
- ***Spark* –** a fast and general compute engine for Hadoop data. Wide range of applications – ETL, Machine Learning, stream processing, and graph analytics.

HDFS

- ***Resilience to hardware failure***
- ***Streaming data access***
- ***Support for large dataset, scalability to hundreds/thousands of nodes with high aggregate bandwidth***
- ***Application locality to data***
- ***Portability across heterogeneous hardware and software platforms***

- ***Single NameNode -*** a master server that manages the file system namespace and regulates access to files by clients.
- ***Multiple DataNodes*** – typically one per node in the cluster. Functions:
  - Manage storage
  - Serving read/write requests from clients
  - Block creation, deletion, replication based on instructions from NameNode


NameNode
Metadata (Name, Replicas)
/home/user1/file1, 3 …
/home/user1/file2, 3 …
HDFS Client
Write
HDFS Client
Read
DataNode
DataNode
DataNode
Replicated

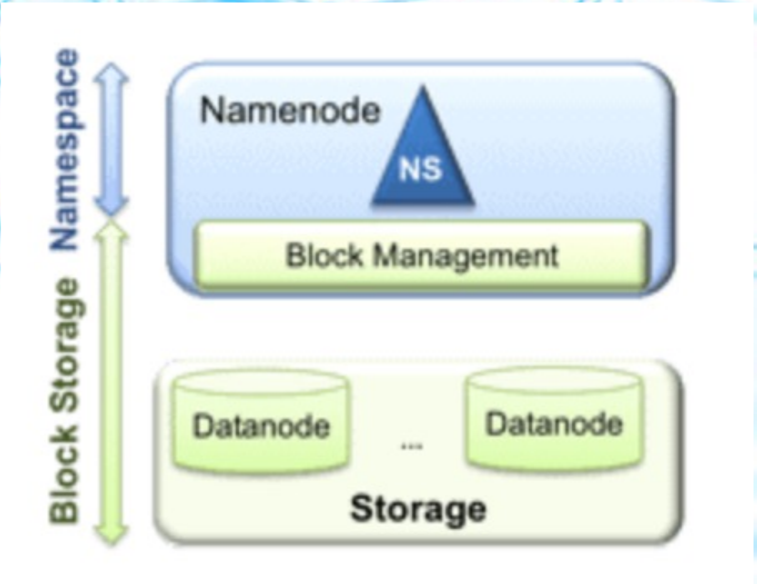
HDFS - Architecture

HDFS has two main layers:

Namespace manages directories, files and blocks. It supports file system operations such as creation, modification, deletion and listing of files and directories.

Block Storage has two parts:

- **Block Management** maintains the membership of datanodes in the cluster. It supports block-related operations such as creation, deletion, modification and getting location of the blocks. It also takes care of replica placement and replication.
- **Physical Storage** stores the blocks and provides read/write access to it.



- **Tight coupling of Block Storage and Namespace**

  Currently the collocation (closeness) of namespace and block management in the namenode has resulted in tight coupling of these two layers. This makes alternate implementations of namenodes challenging and limits other services from using the block storage ***directly***.

- **Namespace scalability**

  While HDFS cluster storage scales horizontally with the addition of datanodes, the namespace does not. Currently the namespace can only be vertically scaled on a single namenode. The namenode stores the entire file system metadata in memory. This limits the number of blocks, files, and directories supported on the file system to what can be accommodated in the memory of a single namenode.

  A typical large deployment at Yahoo! includes an HDFS cluster with 2700-4200 datanodes with 180 million files and blocks, and address ~25 PB of storage. At Facebook, HDFS has around 2600 nodes, 300 million files and blocks, addressing up to 60PB of storage. While these are very large systems and good enough for majority of Hadoop users, a few deployments that might want to grow even larger could find the **namespace scalability limiting.**

- **Performance**

  File system operations are limited to the throughput of a single namenode, which currently supports 60K tasks. The Next Generation of Apache MapReduce will support more than 100K concurrent tasks, which will require multiple namenodes.

- **Isolation**

  At Yahoo! and many other deployments, the cluster is used in a multi-tenant environment where many organizations share the cluster. A single namenode offers no isolation in this setup. A separate namespace for a tenant is not possible. An experimental application that overloads the namenode can slow down the other production applications. A single namenode also does not allow segregating different categories of applications (such as HBase) to separate namenodes.

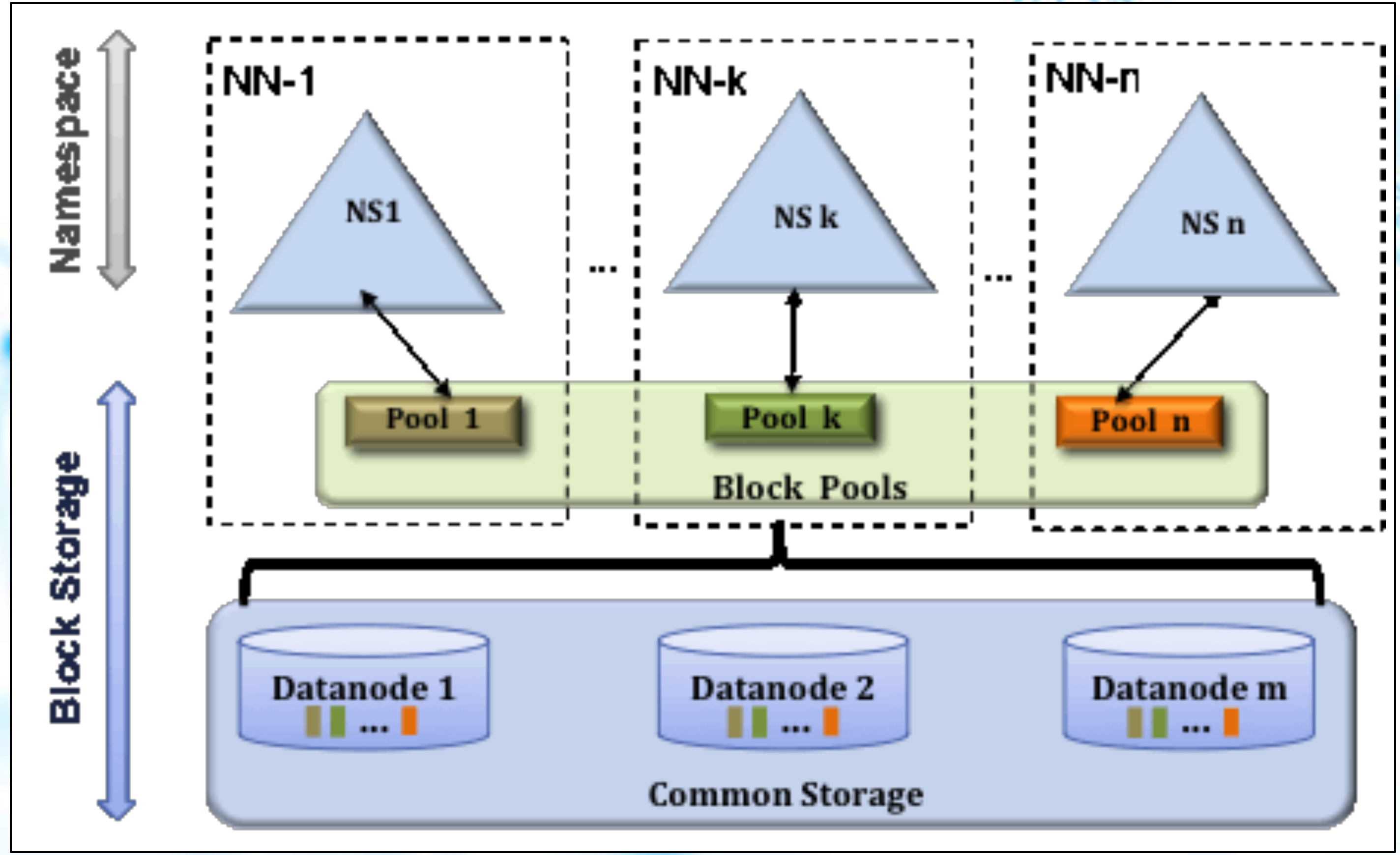
Throughput is a measure of how many units of information a system can process in a given amount of time

- ***HDFS Federation***

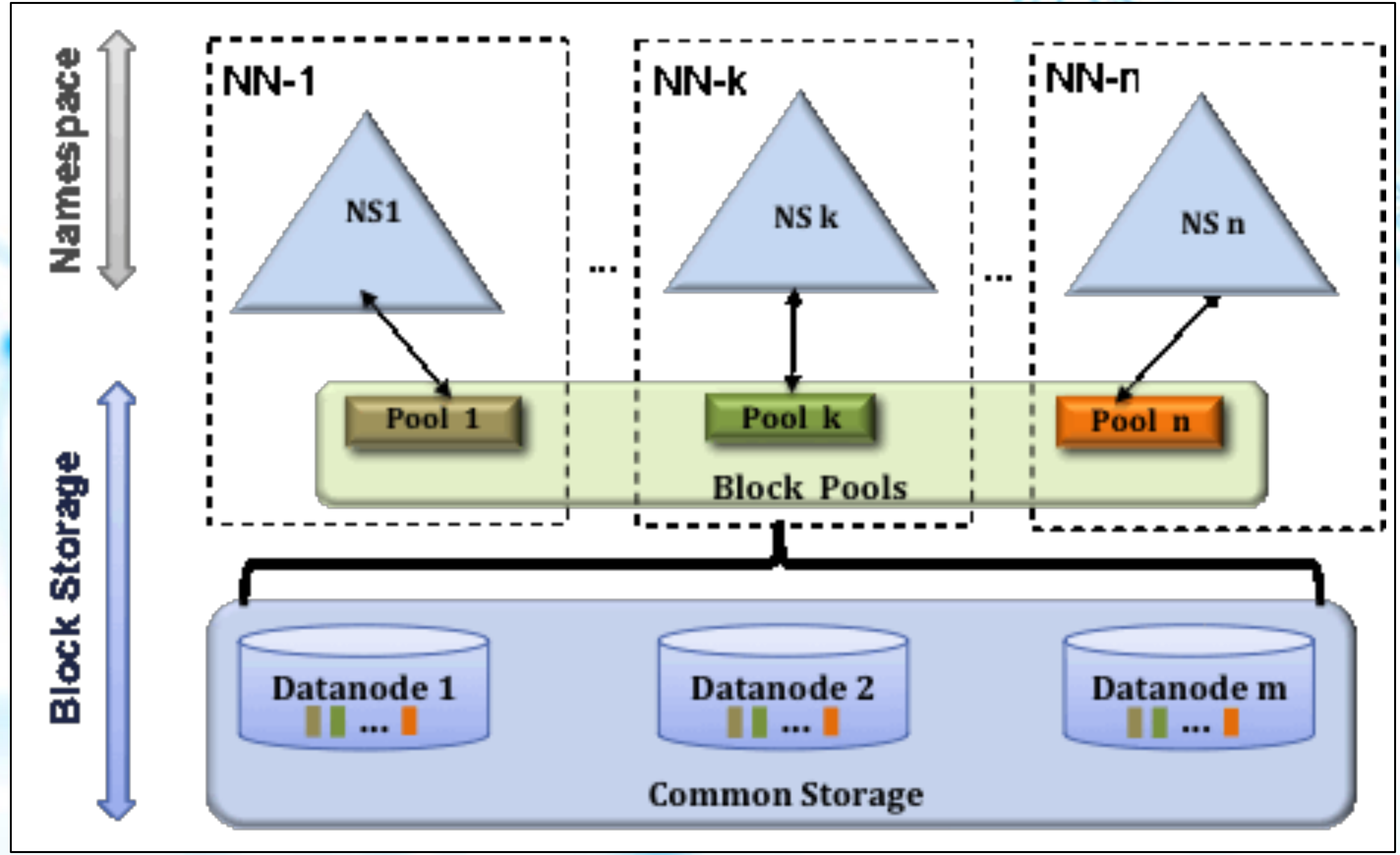
  HDFS Federation improves the existing HDFS architecture through a clear separation of namespace and storage, enabling generic block storage layer. It enables support for multiple namespaces in the cluster to improve scalability and isolation. Federation also opens up the architecture, expanding the applicability of HDFS cluster to new implementations and use cases.
- ***Multiple Namenode servers***
- ***Multiple namespaces***
- ***High Availability – redundant NameNodes***
- ***Heterogeneous Storage and Archival Storage***
  - *ARCHIVE, DISK, SSD, RAM_DISK*

- ***HDFS Federation***

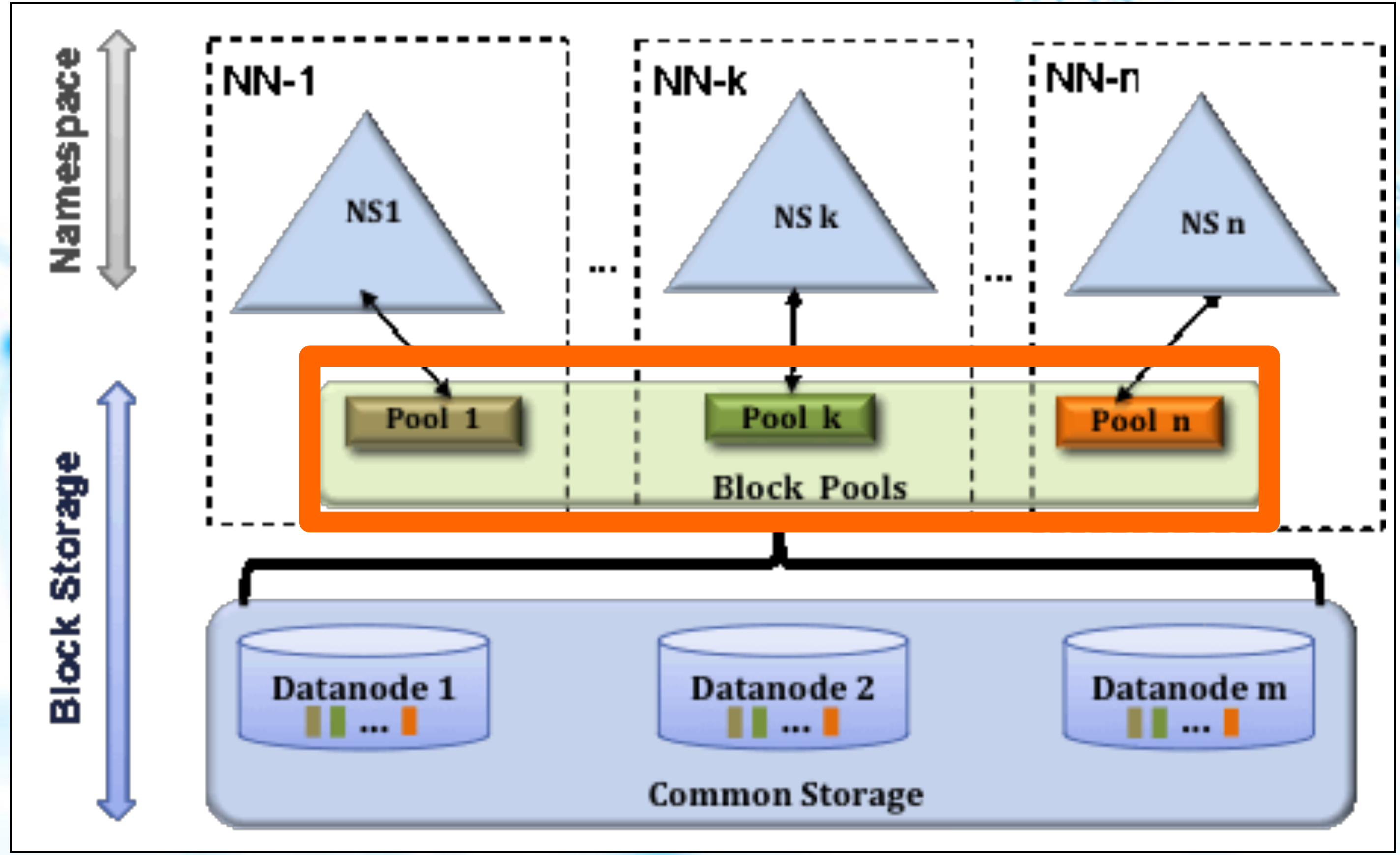
- In order to scale the name service horizontally, federation uses **multiple independent namenodes**/namespaces.
- The namenodes are **federated**, that is, the namenodes are **independent** and don’t require coordination with each other.
- The datanodes are used as common storage for blocks by all the namenodes.
- Each datanode registers with all the namenodes in the cluster.
- Datanodes send periodic heartbeats and block reports and handles commands from the namenodes.



A **Block Pool** is a set of blocks that belong to a single namespace. Datanodes store blocks for all the block pools in the cluster.



A Namespace and its block pool together are called **Namespace Volume**. It is a self-contained unit of management. When a namenode/namespace is deleted, the corresponding block pool at the datanodes is deleted. Each namespace volume is upgraded as a unit, during cluster upgrade.


Namespace
Block Storage
NN-1
NN-k
NN-n
NS1
NS k
NS n
Pool 1
Pool k
Pool n
Block Pools
Datanode 1
Datanode 2
Datanode m
Common Storage

- ***Allows namespace scaling***

- ***Scales up filesystem read/write throughput***

- ***Isolation***

- **Scalability and isolation**

  Support for multiple namenodes horizontally scales the file system namespace. It separates namespace volumes for users and categories of applications and improves isolation.

- **Generic storage service**

  Block pool abstraction opens up the architecture for future innovation. New file systems can be built on top of block storage. New applications can be directly built on the block storage layer without the need to use a file system interface. New block pool categories are also possible, different from the default block pool. Examples include a block pool for MapReduce tmp storage with different garbage collection scheme or a block pool that caches data to make distributed cache more efficient.

- **Design simplicity**

  We considered distributed namenodes and chose to go with federation because it is significantly simpler to design and implement. Namenodes and namespaces are independent of each other and require very little change to the existing namenodes. The robustness of the namenode is not affected. Federation also preserves backward compatibility of configuration. The existing single namenode deployments work without any configuration changes.

  It took only 4 months to implement the features and stabilize the software. The namenode has very few changes. Most of the changes are in the datanode, to introduce block pool as a new hierarchy in storage, replica map and other internal data structures. Other changes include fixing the tests to work with new hierarchy of data structures and tools to simplify management of federated clusters.